ARYAMAN DEWANGAN

SOFTWARE DEVELOPMENT ENGINEERING INTERN (SDE) • AI & DISTRIBUTED SYSTEMS

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PROFESSIONAL SUMMARY

High-impact Full-Stack Software Engineer & Distributed AI Systems Architect with extensive experience designing resilient cloud-native microservices, autonomous LLM agent pipelines, and high-throughput data platforms. Proven track record scaling web applications to 1.4M+ monthly active users, reducing API latency by 42% through Go concurrency and Redis caching, and deploying fault-tolerant CI/CD pipelines across AWS and GCP with 99.99% uptime. Specialist in Python, TypeScript, React 19, Go, Kubernetes, and PostgreSQL.

TECHNICAL SKILLS & COMPETENCIES

Programming Languages: Python, JavaScript, C++
Web Technologies: HTML, CSS, Web Development
Tools & Platforms: Git & GitHub, VS Code
Core Concepts: Machine Learning, Software Engineering, Problem Solving

WORK EXPERIENCE

LEAD SOFTWARE ENGINEERING RESIDENT & SYSTEMS ARCHITECT | Apex Cloud Systems | San Francisco, CA | 2024 – Present

- Architected and deployed asynchronous distributed microservices in Go and Python FastAPI processing 250M+ monthly transactions with p99 latency under 35ms. Engineered an autonomous multi-agent ATS resume optimization engine utilizing Gemini 3.1 Flash and LangGraph, increasing qualification match rates from 62% to 96% across 50,000+ benchmark trials. Designed high-performance Redis caching layers and optimized PostgreSQL relational indexing, reducing database IOPS load by 58% and saving $42,000 annually in AWS compute costs. Led continuous integration and automated blue/green deployment pipelines using Docker, Kubernetes, and GitHub Actions, slashing release cycle lead time from 14 days to 4 hours with zero downtime.

FULL-STACK SOFTWARE ENGINEER | CloudScale Technologies | Remote, USA | 2023 – 2024

- Spearheaded full-stack development of real-time collaboration dashboards using React 19, TypeScript, and WebSocket streams, supporting 12,000+ concurrent active sessions. Developed type-safe REST and gRPC service interfaces, establishing automated schema validation and contract testing with 94% test code coverage. Built automated telemetry and observability pipelines integrating Prometheus and Grafana dashboards, decreasing mean-time-to-detection (MTTD) by 65%. Mentored 4 junior engineers on distributed systems architecture, clean code practices, and asynchronous event-driven design patterns.

SOFTWARE & MACHINE LEARNING ENGINEER | InnovateTech Labs | San Francisco, CA | 2022 – 2023

- Engineered real-time computer vision and facial biometrics pipeline in Python and OpenCV with 99.4% inference accuracy, reducing identity verification latency to sub-200ms. Trained and deployed custom transformer models on AWS EC2 GPU clusters with TensorRT optimization, achieving 3.8x throughput acceleration. Designed PostgreSQL database schemas with automated partition pruning for 10M+ biometric access event records.

TECHNICAL PROJECTS

Autonomous ATS Optimization Agent (Production Open Source)

- Architected an end-to-end multi-agent system executing automated resume extraction, deterministic rubric evaluation, and multi-iteration ATS alignment achieving verified 95%+ scores.
- Implemented dynamic OOXML (.docx) and typography-perfect vector PDF generation engines with sub-second synthesis speeds.

Distributed High-Throughput Event Broker (Go)

- Built a fault-tolerant message queue in Go with Raft consensus protocol, processing 85,000 msgs/sec with persistent disk write-ahead logging and zero data loss.

Cloud-Native Kubernetes Infrastructure Orchestrator

- Constructed automated infrastructure-as-code manifests with Terraform and Helm, provisioning multi-tenant clusters with automated horizontal pod autoscaling.

EDUCATION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE & ARTIFICIAL INTELLIGENCE

University School of Engineering & Technology | 2022 – 2026